

Incident Report

Incident Title

Malware Detection and Persistence Attempt

Date of Detection

Demonstrated by Analyst (Project Demonstration)

Severity Level

Critical

Executive Summary

During a controlled cybersecurity project demonstration, suspicious activity was identified on server1 involving malware execution and persistence attempts. A suspicious user attempted privileged command execution, followed by execution of a malicious binary file named payload.bin. The system identified the file as known malware and terminated the process. The attacker then attempted to create persistence using a system service and initiated outbound communication to a suspicious external IP address.

Affected Assets

- Target Server: server1
- User Account: suspicious_user
- Malware File: /tmp/.hidden/payload.bin
- Persistence Service: persistence.service
- External Suspicious IP: 45.227.10.23

MITRE ATT&CK Mapping

- T1059 – Command Execution
- T1204 – User Execution
- T1547 – Persistence / Autostart Execution
- T1071 – C2 Communication

Detailed Timeline of Events

- 10:18:55 – Suspicious sudo execution detected
- 10:19:04 – Unauthorized binary execution identified
- 10:19:04 – Malware signature matched Trojan.GenericKD.53321
- 10:19:04 – Malicious process terminated
- 10:19:18 – Persistence service file detected
- 10:19:25 – Persistence service enabled
- 10:19:32 – Outbound suspicious connection attempt detected
- 10:19:35 – Firewall blocked suspicious traffic

Impact Assessment

- Malware execution attempt detected
- Persistence mechanism identified
- Suspicious outbound traffic observed
- Potential remote access risk if persistence remained active

Root Cause Analysis

Primary Root Cause:

- Unauthorized malware execution

Contributing Factors:

- Privileged command misuse
- Malware execution from temporary directory
- Persistence through system services

Detection Method

- Splunk log analysis
- Malware signature monitoring
- Persistence monitoring
- Outbound traffic inspection
- SOC dashboard analysis

Actions Performed

- Uploaded malware logs into Splunk
- Investigated suspicious execution activity
- Identified malware and persistence behavior
- Created dashboard and alerts

Recommended Containment Actions

Immediate Actions

- Remove malicious files
- Disable persistence mechanisms
- Block suspicious IP addresses
- Conduct malware validation scans

Long-Term Recommendations

- Deploy EDR monitoring
- Harden privilege controls
- Monitor outbound traffic continuously
- Conduct threat hunting activities

Conclusion

The investigation confirmed malware execution and persistence activity on server1 during a controlled demonstration project. Malware behavior, persistence creation, and suspicious outbound traffic were successfully identified using Splunk monitoring and analysis.